

RGPV | B.Tech First Year

ENGINEERING PHYSICS

(BT-201)

UNIT 5 — Handwritten Notes

Electromagnetic Field Theory

☐ Topics Covered in this Unit

1. *Maxwell's Equations*
2. *Equation of Continuity*
3. *Gauss & Stoke's Theorem*
4. *Gradient & Curl*
5. *Electric Field Intensity*

📖 HOW TO USE THESE NOTES

Blue ink = explanation | Black = headings | Yellow box = exam-important | Green dotted box = trick/mnemonic

UNIT 5

Before we start: the Del Operator ∇

Maxwell's equations are written using vector calculus. One symbol, ∇ ("del" / "nabla"), does three different jobs on a field. Knowing these 3 jobs makes everything else in this unit easy.

$$\nabla = (\partial/\partial x)\hat{i} + (\partial/\partial y)\hat{j} + (\partial/\partial z)\hat{k}$$

Operation	Acts on	Result	Physical meaning
Gradient, $\nabla\phi$	Scalar field ϕ	Vector	Direction & rate of fastest increase
Divergence, $\nabla\cdot\mathcal{F}$	Vector field $\mathcal{F}$	Scalar	Net outflow ("source/sink") per unit volume
Curl, $\nabla\times\mathcal{F}$	Vector field $\mathcal{F}$	Vector	Rotation / circulation per unit area

☞ TRICK

'GDC = Grade, Divide, Curl' — Gradient of Scalar gives vector; Divergence of vector gives Scalar (dot product feel); Curl of vector gives Vector (cross product feel).

1. Maxwell's Equations

DEFINITION

Maxwell's equations are a set of *four fundamental equations* that completely describe classical electricity, magnetism and electromagnetic waves — how electric & magnetic fields are produced by charges, currents, and by each other.

The Four Equations (Differential form)

#	Equation	Based on / Name
<i>I</i>	$\nabla \cdot \mathcal{D} = \rho$	<i>Gauss's Law (Electric)</i>
<i>II</i>	$\nabla \cdot \mathcal{B} = 0$	<i>Gauss's Law (Magnetic) — no monopoles</i>
<i>III</i>	$\nabla \times \mathcal{E} = -\partial \mathcal{B} / \partial t$	<i>Faraday's Law</i>
<i>IV</i>	$\nabla \times \mathcal{H} = \mathcal{J} + \partial \mathcal{D} / \partial t$	<i>Ampere's Law (+Maxwell's correction)</i>

$\mathcal{D}$ = electric flux density, $\mathcal{B}$ = magnetic flux density, $\mathcal{E}$ = electric field, $\mathcal{H}$ = magnetic field, $\mathcal{J}$ = current density, ρ = charge density. ($\mathcal{D} = \epsilon \mathcal{E}$, $\mathcal{B} = \mu \mathcal{H}$ in simple media)

Meaning of each equation (simple words)

- *1st eqn (Gauss-Electric):* Electric field lines start/end on electric charges. Total flux out of a closed surface = charge enclosed.
- *2nd eqn (Gauss-Magnetic):* There is no such thing as an isolated magnetic charge (monopole). Magnetic field lines always form closed loops.
- *3rd eqn (Faraday):* A **changing** magnetic field produces a circulating (curl) electric field — this is the principle behind generators & transformers.
- *4th eqn (Ampere-Maxwell):* A magnetic field is produced both by actual current (*j*) **AND** by a **changing electric field** ($\partial D/\partial t$, called 'displacement current') — Maxwell's own great addition, which predicted EM waves.

Integral form (equivalent, often asked together)

Differential	Integral form
$\nabla \cdot \mathcal{D} = \rho$	$\oint \mathcal{D} \cdot d\mathcal{S} = Q_{\text{enclosed}}$
$\nabla \cdot \mathcal{B} = 0$	$\oint \mathcal{B} \cdot d\mathcal{S} = 0$
$\nabla \times \mathcal{E} = -\partial \mathcal{B} / \partial t$	$\oint \mathcal{E} \cdot d\mathcal{l} = -\partial / \partial t \int \mathcal{B} \cdot d\mathcal{S}$
$\nabla \times \mathcal{H} = j + \partial \mathcal{D} / \partial t$	$\oint \mathcal{H} \cdot d\mathcal{l} = I + \partial / \partial t \int \mathcal{D} \cdot d\mathcal{S}$

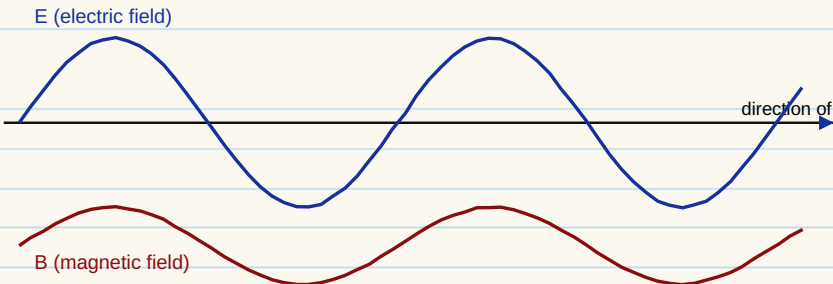


Fig 1.1 — Maxwell's equations predict E and B waves travelling together (EM wave), mutually perpendicular

Significance / Why Maxwell's equations are so important

- They unify electricity, magnetism and optics into a single theory.
- They predict that light itself is an electromagnetic wave, travelling at speed $c = 1/\sqrt{\mu_0 \epsilon_0}$.
- The "displacement current" term (Maxwell's own contribution) fixed a contradiction in Ampere's original law, and made the equations symmetric & consistent.

★ EXAM IMPORTANT

Write all four equations (differential + integral), explain physical significance of each, and derive displacement current / modified Ampere's law — this is the single most repeated long-answer question of the unit (often 7-10 marks).

🔑 TRICK TO REMEMBER

"Electric Diverges, Magnetic never Diverges, Electric Curls from changing B, Magnetic Curls from J and changing D" — read the 4 equations in this story order: $\text{div } \mathcal{D}$, $\text{div } \mathcal{B}$, $\text{curl } \mathcal{E}$, $\text{curl } \mathcal{H}$.

✓ QUICK REVISION

- 4 equations: $\nabla \cdot \mathcal{D} = \rho$, $\nabla \cdot \mathcal{B} = 0$, $\nabla \times \mathcal{E} = -\partial \mathcal{B} / \partial t$, $\nabla \times \mathcal{H} = \mathcal{J} + \partial \mathcal{D} / \partial t$
- Names: Gauss(E), Gauss(B), Faraday, Ampere-Maxwell
- Displacement current $\partial \mathcal{D} / \partial t$ is Maxwell's key addition → predicts EM waves
- Speed of EM wave: $c = 1/\sqrt{\mu_0 \epsilon_0}$

2. Equation of Continuity

DEFINITION

The equation of continuity is the mathematical statement of the **law of conservation of charge**: charge can neither be created nor destroyed, it can only move (flow) from one place to another.

Simple physical picture

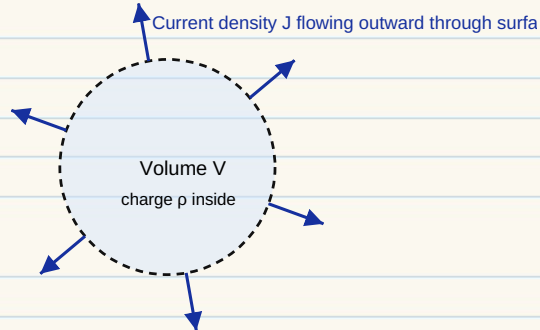


Fig 2.1 — Current flowing out of a closed volume reduces the charge stored inside

- Take a small closed volume V containing charge Q .
- If current flows **out** of this volume through its surface, the charge **inside must decrease** at exactly the same rate (charge cannot vanish, it must have gone out as current).
- "Rate of outflow of current" = "rate of decrease of charge stored"

Derivation (outline)

1. Current flowing out of volume V : $I = \oint \mathbf{J} \cdot d\mathbf{S}$
2. By divergence theorem: $\oint \mathbf{J} \cdot d\mathbf{S} = \int (\nabla \cdot \mathbf{J}) d\tau$
3. Also, $I = -dQ/dt = -\int (\partial \rho / \partial t) d\tau$
4. Equating and cancelling the volume integral (true for any volume):

$$\nabla \cdot \mathbf{J} + \partial \rho / \partial t = 0$$

This is the Equation of Continuity in point (differential) form.

Special case: Steady current (DC, time-independent)

- If current does not change with time, $\partial \rho / \partial t = 0$

$$\nabla \cdot \mathbf{j} = 0$$

This means for steady currents, whatever current flows into a junction/volume must flow out completely — this is exactly **Kirchhoff's Current**

Law (KCL) in circuit theory!

Relation to Maxwell's equations

The equation of continuity is not an independent equation — it can be derived by taking the divergence of the Ampere-Maxwell equation (4th Maxwell eqn) and using the 1st Maxwell equation. This shows that **charge conservation is automatically built into Maxwell's equations**.

★ EXAM IMPORTANT

Derive the equation of continuity from first principles (using divergence theorem), and show how it reduces to $\nabla \cdot \mathbf{j} = 0$ for steady currents. Also "show equation of continuity is consistent with Maxwell's equations" is a common question.

🧠 TRICK TO REMEMBER

"Divergence of $\mathbf{j}$ = minus rate of charge storage" — current spreading OUT (positive divergence) means charge INSIDE is going DOWN. Opposite signs, that's why there's a minus sign!

📝 QUICK REVISION

- Equation of continuity = conservation of charge in mathematical form
- $\nabla \cdot \mathbf{j} + \partial \rho / \partial t = 0$
- For steady current: $\nabla \cdot \mathbf{j} = 0$ ($\equiv$ KCL)
- Derivable from Maxwell's equations $\rightarrow$ charge conservation is built-in

3. Gauss's & Stoke's Theorem

These are two extremely important **vector calculus tools** that let us convert between integrals — and they are exactly what is used to move between the differential and integral forms of Maxwell's equations (Topic 1).

(a) Gauss's Divergence Theorem

STATEMENT

The total outward flux of a vector field $\mathcal{F}$ through a closed surface S equals the volume integral of the divergence of $\mathcal{F}$ over the volume V enclosed by S .

$$\oint_S \mathcal{F} \cdot d\mathbf{S} = \int_V (\nabla \cdot \mathcal{F}) dv$$

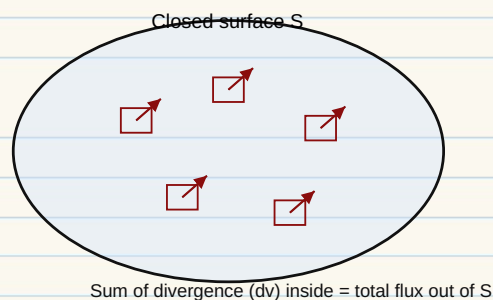


Fig 3.1 — Divergence theorem: surface integral (flux out) = volume integral (sum of sources inside)

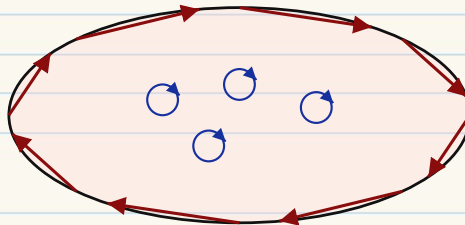
- In simple words: it converts a **surface integral** into a **volume integral** (and vice versa).
- Physically: total flow out through the boundary = sum of all little 'sources' of the field inside.

(b) Stoke's Theorem**STATEMENT**

The line integral of a vector field $\mathcal{F}$ around a closed curve C equals the surface integral of the curl of $\mathcal{F}$ over any open surface S bounded by that curve.

$$\oint_C \mathcal{F} \cdot d\mathbf{l} = \int_S (\nabla \times \mathcal{F}) \cdot d\mathbf{S}$$

Open surface S bounded by closed curve C



Circulation around C = Sum of curl (curl $\mathcal{F}$) over S

Fig 3.2 — Stoke's theorem: circulation around the boundary C = total curl through the surface S

- It converts a **line integral** (around a closed loop) into a **surface integral** (and vice versa).
- Physically: total 'twisting/circulation' around the edge = sum of all the little 'curls' (rotations) on the surface.

Theorem	Converts	Formula	Used to derive
Gauss (Divergence)	Surface $\leftrightarrow$ Volume integral	$\oint \mathcal{F} \cdot d\mathbf{S} = \int (\nabla \cdot \mathcal{F}) dV$	Maxwell eqns I & II, Continuity eqn
Stoke's	Line $\leftrightarrow$ Surface integral	$\oint \mathcal{F} \cdot d\mathbf{l} = \int (\nabla \times \mathcal{F}) \cdot d\mathbf{S}$	Maxwell eqns III & IV

★ EXAM IMPORTANT

State & prove/derive Gauss divergence theorem and Stoke's theorem; also 'use these theorems to convert Maxwell's equations from differential to integral form' is a very frequently asked question linking Topics 1 & 3 together.

🔑 TRICK TO REMEMBER

'**GD-SV, SC-SS**' — Gauss: Divergence converts Surface $\rightarrow$ Volume. Stoke's: Curl converts Surface $\rightarrow$ line/String(loop).

✓ QUICK REVISION

- Gauss theorem: $\oint \mathcal{F} \cdot d\mathbf{S} = \int (\nabla \cdot \mathcal{F}) dV$ (surface $\leftrightarrow$ volume)
- Stoke's theorem: $\oint \mathcal{F} \cdot d\mathbf{l} = \int (\nabla \times \mathcal{F}) \cdot d\mathbf{S}$ (line $\leftrightarrow$ surface)
- Both are used to convert Maxwell's equations between differential and integral forms

4. Gradient & Curl

Gradient (of a Scalar field)

DEFINITION

Gradient of a scalar field Φ is a *vector* that points in the direction of the fastest increase of Φ , and whose magnitude equals the maximum rate of that increase.

$$\nabla\Phi = (\partial\Phi/\partial x)\hat{i} + (\partial\Phi/\partial y)\hat{j} + (\partial\Phi/\partial z)\hat{k}$$

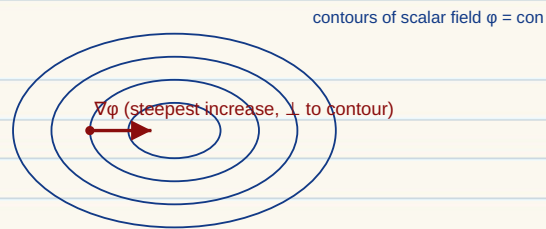


Fig 4.1 — Gradient vector is always perpendicular to the constant- ϕ contour lines, pointing "uphill"

- Example: Φ = temperature in a room. $\nabla\Phi$ points toward the hottest direction from any point.
- In electrostatics: $\mathbf{E} = -\nabla V$ — electric field is the negative gradient of potential.

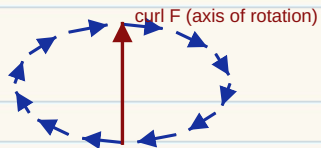
Curl (of a Vector field)

DEFINITION

Curl of a vector field $\mathcal{F}$ is a **vector** that measures the amount & axis of "rotation" or "circulation" of the field at a point.

$$\nabla \times \mathcal{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \mathcal{F}_x & \mathcal{F}_y & \mathcal{F}_z \end{vmatrix}$$

(computed as the determinant of the above 3×3 matrix / cross product of ∇ with $\mathcal{F}$)



Rotating (paddle-wheel) vector field

Fig 4.2 — A field with non-zero curl behaves like a tiny paddle-wheel that would spin if placed in it

- If $\nabla \times \mathcal{F} = 0$ everywhere, the field is called **irrotational** (e.g. electrostatic field $\mathcal{E}$).
- If $\nabla \cdot \mathcal{F} = 0$ everywhere, the field is called **solenoidal** (e.g. magnetic field $\mathcal{B}$).

Operator	Input	Output	Meaning	Example
Gradient $\nabla \phi$	Scalar	Vector	Direction of max increase	$\mathcal{E} = -\nabla \mathcal{V}$
Divergence $\nabla \cdot \mathcal{F}$	Vector	Scalar	Spreading out / source strength	$\nabla \cdot \mathcal{D} = \rho$
Curl $\nabla \times \mathcal{F}$	Vector	Vector	Rotation / circulation	$\nabla \times \mathcal{E} = -\partial \mathcal{B} / \partial t$

★ EXAM IMPORTANT

Physical meaning + formula of gradient, divergence & curl (this comparison table is a favourite short-answer/2-mark question). Also: prove $\text{curl}(\text{grad } \phi) = 0$ and $\text{div}(\text{curl } \mathcal{F}) = 0$ (standard identities, often asked as a quick proof).

🔑 TRICK TO REMEMBER

'Grad walks uphill, Div spreads out, Curl spins around' — three simple actions to remember what each operator physically does.

✓ QUICK REVISION

- Gradient: scalar $\rightarrow$ vector, points in direction of max increase, $\nabla \phi$
- Curl: vector $\rightarrow$ vector, measures rotation, $\nabla \times \mathcal{F}$
- Irrotational: $\nabla \times \mathcal{F} = 0$; Solenoidal: $\nabla \cdot \mathcal{F} = 0$
- $\text{curl}(\text{grad } \phi) = 0$ and $\text{div}(\text{curl } \mathcal{F}) = 0$ (standard identities)

5. Electric Field Intensity

DEFINITION

Electric Field Intensity (E) at a point is defined as the *force experienced by a unit positive test charge placed at that point, without disturbing the source charge.*

$$E = F / q_0 \quad (\text{unit: } \text{N/C or V/m})$$

Field due to a point charge (Coulomb's law based)

$$E = Q / (4\pi\epsilon_0 r^2) \hat{r}$$

Q = source charge, r = distance from charge, $\hat{r}$ = unit vector along the line joining charge to the point, ϵ_0 = permittivity of free space

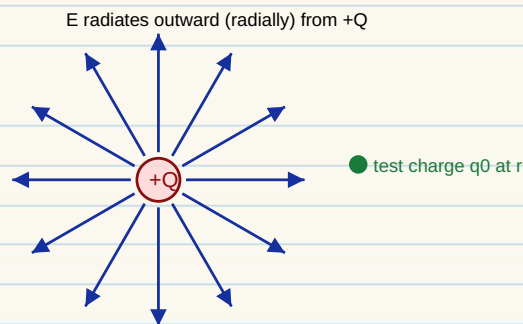


Fig 5.1 — Electric field of a point charge: radially outward for +Q, radially inward for -Q

Key properties

- E is a **vector** quantity (has magnitude and direction).
- Direction: away from a positive charge, towards a negative charge.
- $E \propto 1/r^2$ — field weakens rapidly with distance (inverse square law).
- For multiple charges: total E = **vector sum** of individual fields (superposition principle).

Relation between E and Potential V

$$\mathcal{E} = -\nabla \mathcal{V}$$

Electric field is the negative gradient of electric potential — $\mathcal{E}$ points in the direction of **decreasing** potential (this links Topic 4, Gradient, directly to this topic).

Electric field lines — properties (often asked)

- Field lines start on +ve charges, end on -ve charges (never form closed loops, unlike $\mathcal{B}$).
- They never intersect each other.
- Closer lines = stronger field; farther apart = weaker field.
- They leave/enter a conductor's surface **perpendicular** to it.

Gauss's Law application (quick link)

$$\oint \mathcal{E} \cdot d\mathcal{S} = Q_{\text{enclosed}}/\epsilon_0$$

Used to easily find $\mathcal{E}$ for symmetric charge distributions (sphere, line, sheet) — directly built from Gauss's Divergence theorem (Topic 3) applied to Maxwell's 1st equation (Topic 1).

★ EXAM IMPORTANT

Define $\mathcal{E}$, derive $\mathcal{E}$ for a point charge from Coulomb's law, state properties of field lines, and the relation $\mathcal{E} = -\nabla \mathcal{V}$ — all are commonly asked 2 to 5 mark questions. Numericals finding $\mathcal{E}$ due to point charge / charge distributions are also common.

🔑 TRICK TO REMEMBER

" $\mathcal{E}$ runs downhill from $\mathcal{V}$ " — just like a ball rolls downhill, $\mathcal{E}$ always points from high potential to low potential (that's the minus sign in $\mathcal{E} = -\nabla \mathcal{V}$).

✓ QUICK REVISION

- $\mathcal{E} = \mathcal{F}/q_0$ = force per unit positive test charge
- Point charge: $\mathcal{E} = Q/(4\pi\epsilon_0 r^2)$, direction radial
- $\mathcal{E} = -\nabla \mathcal{V}$ (points toward decreasing potential)
- Field lines: start(+)→end(-), never cross, density shows strength

UNIT 5

📁 Final Formula Sheet & Fast Revision

Topic	Key Formula
Maxwell I (Gauss·E)	$\nabla \cdot \mathcal{D} = \rho$
Maxwell II (Gauss·B)	$\nabla \cdot \mathcal{B} = 0$
Maxwell III (Faraday)	$\nabla \times \mathcal{E} = -\partial \mathcal{B} / \partial t$
Maxwell IV (Ampere-Maxwell)	$\nabla \times \mathcal{H} = \mathcal{J} + \partial \mathcal{D} / \partial t$
Equation of continuity	$\nabla \cdot \mathcal{J} + \partial \rho / \partial t = 0$
Gauss's theorem	$\oint \mathcal{F} \cdot d\mathcal{S} = \int (\nabla \cdot \mathcal{F}) d\mathcal{V}$
Stoke's theorem	$\oint \mathcal{F} \cdot d\mathcal{l} = \int (\nabla \times \mathcal{F}) \cdot d\mathcal{S}$
Gradient	$\nabla \phi, \mathcal{E} = -\nabla \mathcal{V}$
Electric field (point charge)	$\mathcal{E} = Q / (4\pi \epsilon_0 r^2)$
Speed of light	$c = 1 / \sqrt{\mu_0 \epsilon_0}$

🔍 Most Repeated / PYQ Topics (previous year favourites)

1. State & explain all four Maxwell's equations (differential + integral form) with significance
2. Derive displacement current and explain Maxwell's correction to Ampere's law
3. Derive equation of continuity; show it reduces to $\nabla \cdot \mathcal{J} = 0$ for steady current
4. State & prove Gauss's divergence theorem and Stoke's theorem
5. Physical meaning of gradient, divergence and curl (with examples)
6. Electric field intensity due to a point charge; relation $\mathcal{E} = -\nabla \mathcal{V}$

💡 ONE-LINE MEGA MNEMONIC

'Maxwell Conserves charge using Gauss & Stoke's tools, applying Gradient & Curl, to find Electric field' → Maxwell's equations, Continuity, Gauss & Stoke's theorem, Gradient & Curl, Electric field intensity — the order of this whole unit!

All the best for your exam! ✨